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Protein arginine methylation: from enigmatic functions to therapeutic targeting

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Abstract | Protein arginine methyltransferases (PRMTs) are emerging as attractive therapeutic targets. PRMTs regulate transcription, splicing, RNA biology, the DNA damage response and cell metabolism; these fundamental processes are altered in many diseases. Mechanistically understanding how these enzymes fuel and sustain cancer cells, especially in specific metabolic contexts or in the presence of certain mutations, has provided the rationale for targeting them in oncology. Ongoing inhibitor development, facilitated by structural biology, has generated tool compounds for the majority of PRMTs and enabled clinical programmes for the most advanced oncology targets, PRMT1 and PRMT5. In-depth mechanistic investigations using genetic and chemical tools continue to delineate the roles of PRMTs in regulating immune cells and cancer cells, and cardiovascular and neuronal function, and determine which pathways involving PRMTs could be synergistically targeted in combination therapies for cancer. This research is enhancing our knowledge of the complex functions of arginine methylation, will guide future clinical development and could identify new clinical indications.

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Biological methylation is performed by methyltransferases that use *S*-adenosyl-L-methionine (SAM) as a donor to transfer methyl groups to metabolites and biopolymers such as DNA, RNA and proteins. Most protein methylation occurs at the nitrogen atom of basic arginine or lysine side chains or the amino-terminal α-amino group of polypeptides. As a post-translational modification, methylation of arginine, the most basic of all amino acids, results in modification of the guanidinium group, which is protonated at physiological pH. The positively charged guanidinium side chain mediates a wide array of important intramolecular and intermolecular interactions including hydrogen bonding and π-stacking.

Methylation of the guanidinium group of arginine does not change the net positive charge of the residue but distributes the charge over more atoms, and thereby alters the hydrophobicity. Methylation also increases the bulkiness of the side chain and decreases the hydrogen bonding potential[1]. As such, arginine methylation has been shown to regulate binding of methylarginine to protein modules that read this mark, such as plant homeodomain (PHD) zinc fingers, Tudor domains and SH3 domains[2]. The methylation of arginines in histones plays an important role in epigenetic regulation of gene expression[3].

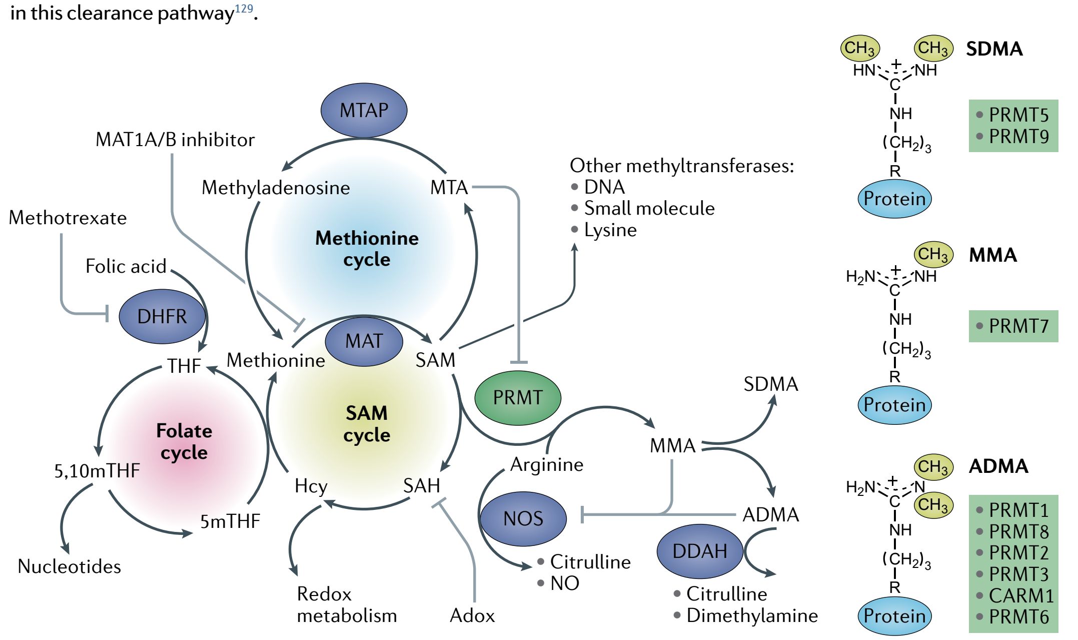
Arginine methylation also weakens the association of arginine-containing proteins with nucleic acids. In particular, RG or RGG sequences have been noted for their tendency to bind RNA through extensive hydrogen bonding with the phosphate backbone or ribose hydroxyls, and through π-stacking and cation–π interactions between the guanidino group and nucleotide bases[4]. RG, RGG and GRG sequences are found in numerous proteins, sometimes in a highly repetitive fashion containing up to 20 repeats[5,6]. Recently, the RG motifs have been implicated in regions of low structural complexity of intrinsically disordered regions (IDRs) that have a propensity to undergo liquid–liquid phase separation or biocondensate formation[7,8]. Biocondensate formation is an organizing principle in the assembly and function of membraneless organelles such as the nucleolus, stress granules and P-bodies, which comprise proteins and RNA and thus regulate transcription, translation and the stress response[9]. RGG and RG motifs in FMR1 and FUS are responsible for recruiting these proteins and their associated RNAs to stress granules and promoting the formation of these membraneless organelles. Arginine methylation of RGG and RG motifs in FMR1 and FUS increases the concentration threshold at which these highly multivalent protein–RNA interactions drive the phase separation-based assembly of organelles[4,10,11]. Another prominent

Box 1 | **Metabolic regulation of arginine methylation**

Arginine methylation by protein arginine methyltransferases (PRMTs) requires *S*-adenosyl-L-methionine (SAM) as a methyl-donating cofactor (see the figure). SAM is at the centre of many biological methylation reactions, from small-molecule metabolites such as catecholamines to RNA, DNA and protein methyltransferases. Thus, SAM metabolism is crucially intertwined with the functions of methyltransferases[254]. The PRMT methylation reaction converts SAM into *S*-adenosyl-L-homocysteine (SAH), which is further converted into L-homocysteine (Hcy). The methyl group regenerating folate cycle provides homocysteine with the vital methyl donor (simplified pathway components shown) from 5-methyl tetrahydrofolate (5mTHF). Dihydrofolate reductase (DHFR), an enzyme in the folate cycle, is a therapeutic target of a cancer drug, methotrexate, that drastically reduces levels of SAM and methylation[255]. Regenerated methionine is part of the methionine cycle in which SAM is synthesized from Met by SAM synthetase (also known as the methionine adenosyltransferase (MAT) complex), a reaction that utilizes ATP. The generation of Met is also linked to the folate cycle, which synthesizes purines. SAM is the methyl donor cofactor for PRMTs to methylate the guanidinium groups of arginine. Distinct PRMTs, listed on the right, perform these methylations.

In addition to generating SAH as a by-product of the methylation reaction, SAM can generate methylthioadenosine (MTA). Both SAH and MTA can inhibit PRMTs in feedback loops; thus, these molecules need to be metabolized. Deficiency in *S*-methyl-5′-thioadenosine phosphorylase (MTAP), the enzyme that converts MTA into methyladenosine, results in the accumulation of MTA and global inhibition of arginine methylation, so PRMTs, including PRMT5, are dependent on MTAP function[145]. A historically used generic compound, adenosine dialdehyde (Adox), inhibits SAH hydrolase and thereby causes excessive SAH accumulation, which globally inhibits cellular methylation.

Circulating monomethyl arginine (MMA) and asymmetrically dimethylated arginine (ADMA) can inhibit the function of nitric oxide synthase (NOS), which generates nitric oxide (NO), a potent vasodilator that regulates blood pressure and inflammation[129]. Symmetrical dimethylarginine (SDMA) is excreted by the kidney, whereas MMA and ADMA are metabolized to citrulline and dimethylamine by dimethylarginine dimethylaminohydrolase (DDAH), the key enzyme in this clearance pathway[129].



category of arginine methylated proteins, rich in RGG and RG motifs, contains splicing-associated factors that also are functionally and physically associated with RNA and DNA (discussed below).

Protein arginine methylation has been implicated in numerous diseases. Because protein methylation is a targetable modification, its therapeutic potential has been investigated in preclinical models and is being tested in the clinic for oncology indications. The growing number of other potential indications linked to arginine methylation stems from the complex biology of the protein arginine methyltransferase (PRMT) enzymes and their downstream substrates. At the same time, some of the downstream biological processes controlled by arginine methylation may be therapeutic liabilities. In this Review, we discuss the breadth and complexity of this system and suggest potential future directions for exploring arginine methylation in drug discovery.

π-Stacking
Non-covalent bonding between aromatic moieties that can align themselves in a parallel or orthogonal fashion.

Liquid–liquid phase separation or biocondensate
A reversible molecular interaction driving condensation of macromolecules into a dense phase resembling liquid droplets and existing within a dilute phase.

Physiology of arginine methylation

PRMT enzymes. Concurrent with the discoveries of the RG/RGG/GRG-rich motifs and their propensity for binding RNA, the profiling of methylated proteins revealed that arginine methylated RG/RGG motifs are well represented in the proteome[6] and, in fact, the RG/RGG motif is strongly preferred by PRMT enzymes[12–16]. There are at least nine different PRMTs that monomethylate and dimethylate arginines within proteins using SAM as a cofactor (BOX 1). Several excellent, in-depth reviews discuss each of these PRMT enzymes in more detail[3,17–19]. PRMTs are divided into three types: the type I enzymes PRMT1, PRMT2, PRMT3, CARM1 (also known as PRMT4), PRMT6 and PRMT8 dimethylate their

Ribonucleoprotein
(RNP). A protein complex with ribonucleic acids to influence processing, localization and stability of the complex.

Small nuclear ribonuclear protein complexes
(snRNPs). Complexes of proteins and small non-coding RNAs of 100–600 nucleotides that are essential in transcription, pre-mRNA splicing and processing, and transcription termination.

substrates asymmetrically (with both methyl groups on one of the guanidino nitrogen atoms). Type II enzymes PRMT5 and PRMT9 are symmetrical dimethyltransferases (which transfer one methyl group to each of the two terminal guanidino nitrogen atoms). The sole member of the type III group is PRMT7, which can only monomethylate arginines (BOX 1). Interestingly, mass spectrometry studies indicate that PRMT1 and PRMT5 may also monomethylate their substrates[14,16]; however, this observation requires further studies, especially given the complex regulation of PRMT1 processivity[20] and the inherently distributive nature of the PRMT5 enzyme[21]. Such studies are also needed to shed light on the number of potential methylation states induced by a single PRMT. The overlapping substrate specificity and substrate competitive behaviour of PRMTs are further illustrated by the phenomenon of substrate scavenging, in which the absence of one enzyme leads to methylation of the substrate by another enzyme; PRMT1 and PRMT5 scavenge each other's substrates[22]. Overall, only a small proportion of all known arginine methylated protein substrates have been definitively assigned to an individual PRMT[23]. Thus, the full extent of methylation and the functional importance of arginine methylation on many proteins remain to be uncovered.

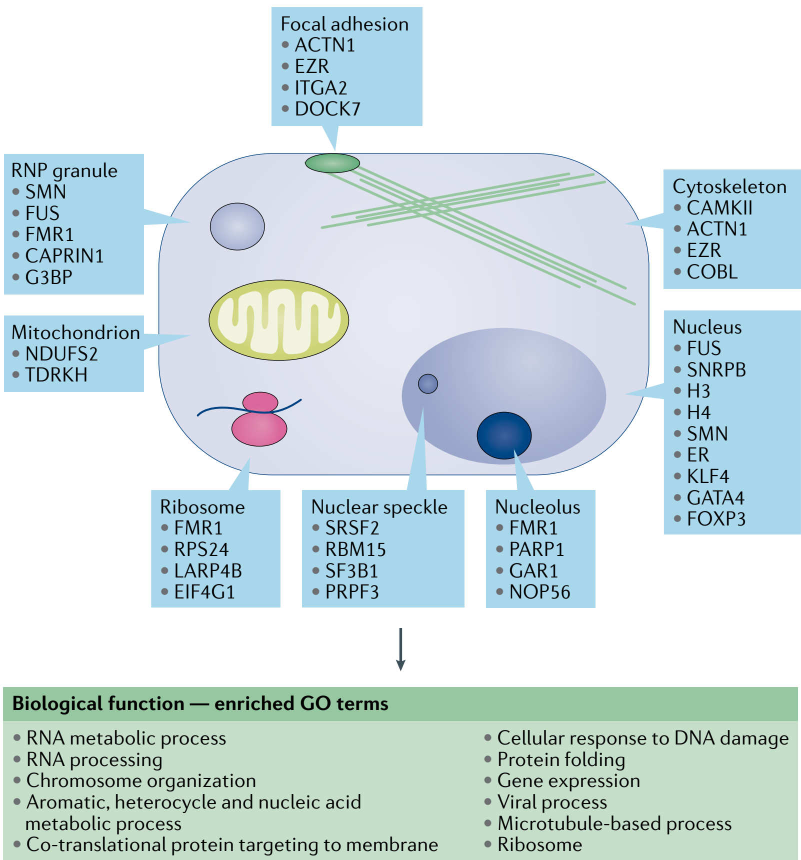


Fig. 1 | **Distribution and functions of arginine methylated proteins.** Arginine methylated proteins (4,087 unique proteins) from PhosphoSitePlus[23] were analysed for the cellular component and biological function Gene Ontology (GO) term enrichment using EnrichR GO term analysis[266] and the most significant terms are summarized. Representative proteins from different cellular compartments are shown. ACTN1, α-actinin 1; CAMKII, Ca^{2+}/calmodulin-dependent protein kinase 2; COBL, protein cordon-bleu; DOCK7, dedicator of cytokinesis 7; EIF4G1, eukaryotic initiation factor 4G1; ER, oestrogen receptor; EZR, ezrin; FOXP3, forkhead box P3; G3BP, Ras GTPase-activating protein-binding protein; H3, histone H3; ITGA2, α2 integrin; KLF4, Kruppel-like factor 4; LARP4B, La-related protein 4B; NDUFS2, NADH dehydrogenase [ubiquinone] iron–sulfur protein 2; NOP56, nucleolar protein 56; PARP1, poly(ADP-ribose) polymerase 1; PRPF3, pre-mRNA processing factor 3; RBM15, RNA-binding protein 15; RNP, ribonucleoprotein; RPS24, 40S ribosomal protein S24; SF3B1, splicing factor 3B1; SMN, survival of motor neuron; SNRPB, small nuclear ribonucleoprotein-associated proteins B and B1; SRSF2, serine/arginine-rich splicing factor 2; TDRKH, Tudor and KH domain-containing protein.

Consequences of arginine methylation. There are more than 4,000 reported arginine methylated proteins in PhosphoSite[23]. The functional analysis of these methylated proteins implicates a large proportion of them in processes associated with RNA metabolism, such as RNA biogenesis, processing, splicing and gene expression, as well as in chromatin organization, translation, and metabolic and microtubule-associated processes (FIG. 1). Accordingly, these arginine methylated proteins are distinctly localized in the nucleus, nucleolus, ribosome, ribonucleoprotein (RNP) granules, microtubules and focal adhesions. Although it is unlikely that arginine methylation determines the entire functional repertoire of these proteins, this modification modulates protein functions. For example, arginine methylation is relatively well studied in RNA-binding proteins (RBPs), which play critical roles in the RNA-associated processes of splicing, translation or RNA stability regulation. Often, methylated arginines are located in protein–protein or protein–nucleic acid interaction interfaces where the methylation event regulates the strength of these interactions and, thus, the assembly of macromolecular complexes[24] (extensively reviewed elsewhere[3,19]). Interestingly, some little-studied arginine methylated proteins are localized to focal adhesions and the cytoskeleton — which suggests that they may be involved in microtubule-based processes and signalling rather than RNA regulation. Although this extensive list of functionally diverse substrates for PRMTs illustrates the potential range of regulatory repertoire and downstream biological functions of PRMTs, it is possible that many more arginine methylated proteins exist. The majority of arginine methylation profiling studies have used immunoaffinity methods with antibodies that preferentially identify RG/RGG methylated motifs[15]; thus, new methodology is needed to elucidate the full extent of cellular protein arginine methylation.

Several highly arginine methylated proteins are associated with splicing. PRMT5 methylation of Sm proteins (a type of RBP) enables them to bind to the Tudor domain of survival motor neuron (SMN), which promotes the assembly of functional small nuclear ribonuclear protein complexes (snRNPs) that are essential for spliceosome function[25]. PRMT5 also methylates serine/arginine-rich splicing factor 1 (SRSF1) to promote efficient splicing[26]. The absence or inhibition of PRMT5 thus reduces spliceosome assembly, and leads to exon skipping and retention of introns with weak 5′ splice donor sites[27,28]. Splicing programme regulation could diversify the transcriptome, and several specific splicing factor programmes are at play in development, differentiation and cancer[29].

Development of tissue-specific processes. In humans, arginine methyltransferase mutations have been reported for *PRMT7*, for which germline heterozygous or homozygous mutations result in short stature, brachydactyly, intellectual developmental disability and seizures (SBIDDS) syndrome[30]. There are no cancer-associated mutations in PRMTs, although the levels of some PRMTs are increased in cancer[31]. In mice, genetic disruption of *Prmt1* or *Prmt5* is embryonically lethal (for reviews see REFS[18,32]). Many arginine methylated proteins have been associated with the regulation of embryogenesis, neurogenesis and specific T cell functions, thus providing possible links to roles in developmental processes and tissue-specific diseases (FIG. 1; TABLE 1).

The tissue-specific functions of arginine methylation derive from a combination of spatio-temporal variations in the expression of PRMTs and their substrate proteins (TABLE 1), which are potentially important in considering side effects of PRMT-targeting therapies. For example, in the central nervous system, methylation driven by PRMT1 and PRMT5 is essential for myelination and oligodendrocyte differentiation, and PRMT8, which is a neuron-specific, membrane-localized methyltransferase, plays a crucial role in neuronal differentiation; genetic disruption of PRMT8 results in a defect in motor coordination and hyperactivity[33–36]. Methylation of signalling receptors such as platelet-derived growth factor receptor (PDGFR) by PRMT5 safeguards activated forms of this receptor from recognition and degradation by the ubiquitin–proteasome pathway. The methylation of PDGFR promotes oligodendrocyte precursor growth and self-renewal, leading to appropriate axon myelination[37], and may highlight a therapeutic target to modulate PDGFR levels using PRMT-targeting compounds that can penetrate the blood–brain barrier. Several other signalling molecules and receptor systems, such as SMAD–TGFβ and epidermal growth factor receptor (EGFR), contain methylated arginines that may contribute to their roles in development[3,18,38].

Arginine methylated proteins that are involved in developmental regulation are often localized in mitochondria. Apart from the nine well-studied PRMTs, a mitochondrial arginine methyltransferase, NDUFAF7 (also known as MidA), is necessary for mitochondrial respiration complex I assembly and could be important in methylating multiple mitochondrial proteins. Because its known substrate, NADH dehydrogenase [ubiquinone] iron–sulfur protein 2 (NDUFS2), is linked to several neurodevelopmental diseases[39,40], it is possible that NDUFAF7 contributes to mitochondrial proteome methylation and the regulation of neuronal energy metabolism.

PRMTs contribute to homeostasis in bone and cartilage[41–43]. PRMTs regulate bone formation and resorption, which are carried out by osteoblasts and osteoclasts, respectively. For example, PRMT3 epigenetically increases the expression of microRNA (miR)-3648 and thus promotes the osteogenic differentiation of mesenchymal stem cells and bone remodelling[44]. Genetic or pharmacological inhibition of PRMT3 reduces bone mass in mice, indicating that this enzyme could have diagnostic and therapeutic roles in osteoporotic diseases[44]. Both PRMT1 and PRMT5 are indispensable for osteoclast differentiation and bone resorption because they regulate nuclear factor-κB (NF-κB) activity, providing a potential therapeutic strategy for osteoclast-related disorders by inhibiting the aberrant activation of PRMT1 or PRMT5 (REFS[41,42,45]).

PRMTs play a key role in cardiomyocyte function in both healthy and pathological conditions[46–50]. PRMT5 is required for normal cardiac function, as it inhibits the transcriptional activity of GATA4 and the expression of hypertrophic genes. Likewise, PRMT1 prevents hyperactivation of Ca^{2+}/calmodulin-dependent protein kinase 2 (CAMKII), through methylation, to preserve cardiovascular homeostasis. In vivo, cardiac-specific ablation of PRMT1 results in a rapid progression to heart failure with dilated cardiomyopathy; these mice exhibit aberrant cardiac alternative splicing[48,50], highlighting the protective role of PRMTs in the cardiac system.

In contrast to the positive roles of PRMT1 and PRMT5 in cardiac tissue function, CARM1 levels are increased in ischaemic hearts and hypoxic cardiomyocytes[51]. The overexpression of CARM1 in mice inhibits proliferation and induces apoptosis in cardiomyocytes, leading to reduced survival[51]. Future studies utilizing pharmacological selective small-molecule modulators will ascertain whether CARM1 inhibition would be beneficial in cardiovascular diseases. Thus, the long-term inhibition of PRMTs may disrupt regeneration and/or homeostasis of cardiac tissues and should be considered when therapeutically targeting PRMTs.

Arginine methylation in disease

Collectively, arginine methylated proteins, through their involvement in a wide array of biological functions, have been linked to numerous diseases (FIG. 1; TABLE 1). These links provide only tentative connections, as arginine methylation does not necessarily modulate disease-essential functions of these proteins. However, they offer further avenues for investigating the functional consequences of methylation for many of these proteins where no information is available other than the occurrence of methylation. In this section, we discuss the known and potential roles of arginine methylated proteins and PRMTs in the disease categories that have been linked to arginine methylation, such as cancer and neurological and inflammatory diseases, and indicate possible unexplored links as novel avenues for further biological and therapeutic exploration.

Cancer and arginine methylation. Arginine methylation is well studied in cancer, and most PRMTs have been implicated in the cancer-associated regulation of epigenetics and chromatin, transcription, signalling, RNA metabolism and DNA repair[3,18,32]. Consistently, arginine methylation of several proteins is prevalent in breast, lung and colon cancers and leukaemia, and may play important roles in these diseases (TABLE 1).

The early studies on histone methylation uncovered an intricate interplay between histone H3 trimethylated at lysine 4 (H3K4me3), a transcriptional activation mark, and H3 symmetrically methylated at arginine 2 (H3R2me2s). H3R2me2s methylation is performed by PRMT5 and recognized by WD repeat-containing protein 5 (WDR5), a component of the mixed lineage leukaemia (MLL) complex, which drives methylation of H3K4 and activates transcription of leukaemia-sustaining HOX genes[52,53]. In contrast, H3 asymmetrically dimethylated at arginine 2 (H3R2me2a) by PRTM6 prevents WDR5 from binding chromatin and activating gene transcription[54]. In solid cancers, this crosstalk is especially relevant because HOX genes are implicated in epithelial–mesenchymal transition (EMT), which promotes tumour growth and metastasis[55].

In addition, PRMT1 is found with MLL translocation-associated transcriptional complexes, which drive leukaemogenic gene expression programmes[56]; these complexes also rely on CARM1 activity[57]. Asymmetrical dimethylation of H4R3 at the *ZEB1* promoter by PRMT1 leads to ZEB1 overexpression, which drives EMT and increases migration and invasion of breast cancer cells[58]. Histone arginine methylation drives another transcriptional programme that is essential to breast cancer: PRMT5 methylates H3R2 in the promoter of *FOXP1*, and forkhead box protein P1 (FOXP1) controls breast cancer stem cell self-renewal and proliferation[59].

Gene regulatory programmes can also be modulated by direct transcription factor methylation. Methylation of Kruppel-like factor 4 (KLF4) by PRMT5 reduces its affinity for von Hippel–Lindau disease tumour suppressor (VHL), resulting in increased stability of KLF4, overexpression of its downstream genes and breast cancer progression[60]. The importance of arginine methylation in the cancer-associated *MYC* oncogene transcriptional programme is exemplified by MYC-dependent lymphoma and breast cancer, in which high levels of PRMT5 (a downstream MYC target) impose a splicing programme that maintains the cancer transcriptome[61,62]. In the Eμ-MYC transgenic mouse model of lymphoma, PRMT5 is required for tumour maintenance[62]. The roles of MYC and PRMT5 are generally well established in lymphoma, as PRMT5 levels are upregulated in patient samples and cell lines[63]. MYC also regulates translation, and several studies point to the role of arginine methylation in protein translation control[3,19,64]. Thus, the coordinated regulation of transcription, splicing and translation fuelled by oncogenic MYC or other transcription factors could potentially be targeted by modulating arginine methylation.

Splicing programme alterations are widespread in cancer[65,66]. Indeed, mutations in splicing factors found in leukaemia and myelodysplastic syndrome (MDS) are haploinsufficient, rendering cells vulnerable to further splicing inhibition via PRMT modulation. Inhibition of PRMT5 and type I PRMTs in leukaemia with splicing factor mutations induces a tumour-suppressive response in animal models[14]. Furthermore, PRMT5 methylation of SRSF1 is a driver of leukaemic splicing programmes and is therefore a vulnerability for MLL-rearranged leukaemia[26]. PRMT1 methylation of RNA-binding protein 15 (RBM15) destabilizes this RBP, which regulates splicing of genes critical to megakaryocytic differentiation; PRMT1 overexpression blocks differentiation and promotes leukaemogenesis[67]. Overall, strong links to splicing deregulation in leukaemia and MDS, elevated levels of PRMT5 in lymphoma and extensive haematological malignancy transcriptional programme regulation by PRMTs[68] have provided a rationale to target PRMT5 and PRMT1 in blood cancers. Targeting splicing via PRMT5 inhibition has also been demonstrated in solid tumours[69]. For example, widespread alterations in splicing were noted in a panel of patient-derived glioblastoma stem cells upon PRMT5 inhibition, with resultant defects in cell cycle machinery and stemness. In this study, the differential responses among patient samples to PRMT5 inhibition was related to differential pre-existing splicing signatures, further emphasizing the role of PRMT5 in splicing biology[70].

Myelodysplastic syndrome (MDS). A complex group of disorders defined by peripheral cytopenia, dysplastic haematopoietic progenitors affecting particular lineages, a hypercellular or hypocellular bone marrow and a high risk of progression to leukaemia.

PRMTs also function as co-regulators of nuclear receptors that mediate hormone-dependent gene regulation[71]. In breast cancer, PRMT1 methylates the arginine residue within the DNA-binding domain of oestrogen receptor-α (ERα) in vitro and in vivo. This methylation event triggers its interaction with the tyrosine kinase SRC, the p85 subunit of phosphatidylinositol 3-kinase (PI3K) and focal adhesion kinase (FAK), leading to the activation of key downstream signalling kinases such as MAPK and AKT[72,73]. As activation of these pathways might contribute to endocrine resistance, further investigations are required to determine the possibility of using PRMT inhibitors to sensitize resistant breast cancer cells to ER-targeted therapies. In prostate cancer, PRMT5 transcriptionally activates androgen receptor (AR) to promote cell growth. Thus, inhibition of arginine methylation impairs the growth of xenograft tumours in mice models that express AR[74].

Methylome analyses have also identified that 3% of metabolic enzymes are arginine methylated, implying that PRMTs could modulate nutrient sensing and metabolic regulation during tumorigenesis[15,75]. Methylated malate dehydrogenase (MDH1) suppresses mitochondrial respiration and glutamine metabolism in pancreatic cancer cell lines, and thus sensitizes these cells to oxidative stress[76]. CARM1-mediated methylation of glyceraldehyde-3-phosphate dehydrogenase (GAPDH) signals glucose availability and slows down glycolysis in an AMP-activated protein kinase (AMPK)-dependent manner in liver cancer cell lines[77]. Thus, methylated GAPDH inhibits the growth of xenograft tumours in mice, suggesting that small molecules that function as CARM1 activators might have therapeutic potential for suppressing liver cancer growth[77]. Interestingly, arginine methylation of 6-phosphofructo-2-kinase/fructose-2,6-bisphosphatase 3 (PFKFB3) by PRMT1 increases glycolytic flux, whereas its hypomethylated form repurposes glucose for the pentose phosphate pathway, thereby regulating oxidative stress[78]. Despite these advances in our understanding, the biological function of most methylated metabolic enzymes remains unknown.

In addition to metabolic enzyme modulation and methylation, PRMTs also methylate chaperone proteins that play critical roles in cancer signalling, proteostasis and drug resistance. Heat shock protein 70 (HSP70) and HSP90, key regulators in cancer, are arginine methylated[79,80]. HSP70 methylation by PRMT1 and PRMT7 regulates apoptosis, cancer aggressiveness, drug sensitivity and transcription factor signalling[81–83].

Table 1 | **Tissue-specific roles of PRMTs in disease**

PRMT	Mechanism (target)	Cellular roles	Disease
Lung			
PRMT1	NA	Cell migration	Idiopathic pulmonary fibrosis[122]
PRMT1	Altered expression (eotaxin 1; CCR3)	Inflammation response	Asthma[121]
PRMT1	Altered expression (PGC1α; HSP60)	Mitochondrial mass	Asthma[120]
PRMT5	Altered expression (EMT genes)[55]	Metastasis	Lung cancer
	Altered expression (SHARPIN)[184]		
	Altered expression (miR-99 family/FGFR3)[185]		
PRMT7	Interaction with carcinogenesis associated factors (HSPA5/EEF2)[186]		
Liver			
PRMT1	Altered expression (PGC1α)[187]	Lipogenesis	Fatty liver diseases
PRMT3	Altered cofactor binding (LXR)[188]	Lipogenesis	
PRMT5	Altered expression (PPARα; PGC1α)[189]	Mitochondrial biogenesis	
PRMT1	Methylation (HDAg)[112]	Viral RNA replication	Hepatitis delta virus infection
PRMT1	Altered cofactor binding (HBx)[190]	Viral transcription	Hepatitis B virus infection[a]
CARM1	Methylation (GAPDH)[77]	Glucose metabolism	Liver cancer
PRMT5	Methylation (SREBP)[191]	Lipogenesis	
	Altered expression (BTG2)[192]	ERK signalling	
	PRMT5 activity (LINC01138)[193]	PRMT5 degradation	
	Altered cofactor binding (LYRIC)[194]	WNT–β-catenin signalling	
PRMT6	Methylation (CRAF)[195]	RAS–RAF–MEK signalling	Liver cancer[a]
Pancreas			
PRMT1	Methylation (GLI1)[196]	Oncogenic activity	Pancreatic cancer
PRMT3	Methylation (hnRNPA1)[169]	ABCG2 expression	
CARM1	Methylation (MDH1)[76]	Glutamine metabolism	
PRMT5	Altered stability (MYC)[197]	Glucose metabolism	
Muscle			
PRMT1	Altered expression (PRMT6)[198]	FOXO3 signalling	Muscle atrophy
	Methylation (EYA1)[199]	Stemness	
CARM1	Altered cofactor binding (UPF1)[200]	mRNA decay	
	Methylation (splicing factors)[201]	mRNA splicing	
Bone			
PRMT1	Altered cofactor binding (NF-κB)[41]	NF-κB signalling	Osteoporosis
PRMT5	Altered expression (interferon genes)[42,45]	Type I interferon signalling	
PRMT1	Impaired formation of the translation initiation complex[202,203]	Translation-related pathways	Osteosarcomas
PRMT6	Altered expression (TSP1)[204]	Cell migration	
Prostate			
PRMT5	Altered methylation (AR)[160]	ERG signalling	Prostate cancer
Ovary			
PRMT1	Altered H4R3 methylation via binding and phosphorylation (DNA-PK)[167]	DNA damage response	Ovarian cancer
	Methylation (FAM98A)[205]	NA	
CARM1	Methylation (SMARCC1)[206]	EZH2 signalling; DNA damage response	
PRMT5	Altered expression (E2F1)[207]	NA	
Type I PRMTs	ADMA production[208]	NO signalling	Polycystic ovary syndrome
Brain			
PRMT1	Methylation (FUS)[90,91,209]	Phase separation	Amyotrophic lateral sclerosis
PRMT5	Methylation (FUS)[210]	Phase separation	
PRMT8	Not known[211]	Not known	Behavioural defects[a]

Table 1 (cont.) | **Tissue-specific roles of PRMTs in disease**

PRMT	Mechanism (target)	Cellular roles	Disease
Brain (cont.)			
PRMT1	Methylation (KCNQ)[212]	Channel activity	Neuronal excitability[a]; epileptic seizures[a]
PRMT7	Methylation (NALCN)[213]	Channel activity	
PRMT5	Altered expression (E2F1)[104]	GSK3β–NF-κB signalling	Alzheimer disease[a]
PRMT5	Altered cofactor binding (HTT)[101]	HTT toxicity	Huntington disease[a]
PRMT5	Altered expression (PTEN)[214]	AKT–ERK signalling; cell cycle progression	Glioblastoma
	Altered RNA splicing (oncogenic genes)[28]	Cell cycle progression, apoptosis, senescence	
	Methylation (hnRNPA1)[175]	mTOR signalling	
	Altered expression (RNF168)[215]	DNA damage response	
	Altered expression (LRP12)[216]	Cell cycle progression, apoptosis, migration	
Heart			
PRMT1	Methylation (CAMKII)[50]	Kinase signalling	Cardiomyopathy[a]
	Altered RNA splicing (EIF4A2)[48]	Translational activity	
PRMT5	Methylation (GATA4)[47]	Transcriptional activity	
Kidney			
Type I PRMTs	ADMA production[217–219]	NO signalling	Chronic kidney disease
CARM1	Altered expression (AMPKα; CB1R)[220]	Notch1 signalling	Diabetic nephropathy[a]
Breast			
PRMT1	Methylation (ASK1)[221]	Apoptosis	Breast cancer
	Methylation (BRCA1)[222]	DNA damage response	
	Methylation (C/EBPα)[223]	Cyclin D1 signalling	
	Methylation (ERα)[72]	IGF1 signalling	
	Methylation (EGFR)[224]	EMT signalling	
	Altered expression (ZEB1)[58]		
PRMT2	Altered cofactor binding (ERα)[225,226]		
CARM1	Methylation (SMARCC1)[227]	Metastasis	
	Methylation (LSD1, BAF155)[227,228]	Metastasis	
	Methylation (MED12)[229]	TGFβ signalling	
	Altered cofactor binding (ERα)[230]	Cell differentiation	
PRMT6	RNA splicing (growth factors)[231]	NA	
	Altered cofactor binding (UHRF1)[232]	DNA demethylation	
PRMT5	Altered expression (OCT4; KLF4; c-MYC)[233]	Stemness	
	Altered expression (FOXP1)[59]	Stemness	
	Altered expression (EMT genes)[55]	Invasion and metastasis	
	Methylation (KLF4)[60]	Cell cycle	
	Methylation (ZNF326)[234]	Migration	
	Methylation (PDCD4)[235]	NA	
PRMT7	Altered expression (EMT genes)[236]	Invasion and metastasis	
	Methylation (SHANK2)[237]	FAK signalling, metastasis	
	Automethylation[238]	EMT signalling	

Most of the diseases listed here have the potential to be treated with PRMT inhibitors. Those that are potential therapeutic liabilities for treatment with PRMT inhibitors are indicated. ADMA, asymmetrical dimethylarginine; AMPK, AMP-activated protein kinase; AR, androgen receptor; ASK1, apoptosis signal-regulating kinase 1; BRCA1, breast cancer type 1 susceptibility protein; CAMKII, Ca^{2+}/calmodulin-dependent protein kinase II; CB1R, cannabinoid receptor 1; C/EBPα, CCAAT/enhancer-binding protein-α; DNA-PK, DNA-dependent protein kinase; E2F1, E2 promoter binding factor 1; EEF2, elongation factor 2; EGFR, epidermal growth factor receptor; EIF4A2, eukaryotic initiation factor 4A-II; EMT, epithelial–mesenchymal transition; ERα, oestrogen receptor-α; ERG, ETS-related gene; EYA1, eyes absent homologue 1; FAK, focal adhesion kinase; FGFR3, fibroblast growth factor receptor 3; FOXP1, forkhead box protein P1; GAPDH, glyceraldehyde-3-phosphate dehydrogenase; GSK3β, glycogen synthase kinase-3β; HBx, hepatitis B protein X; HDAg, hepatitis D antigen; hnRNPA1, heterogeneous nuclear ribonucleoprotein A1; HSP60, heat shock protein 60; HTT, huntingtin protein; IGF1, insulin-like growth factor I; KCNQ, potassium voltage-gated channel subfamily KQT; KLF4, Kruppel-like factor 4; LINC01138, long non-coding RNA 01138; LRP12, low-density lipoprotein receptor-related protein 12; LSD1, lysine-specific histone demethylase; LXR, liver X receptor; MDH, malate dehydrogenase; MED12, mediator of RNA polymerase II transcription subunit 12; miR, microRNA; NA, not applicable; NALCN, sodium leak channel non-selective protein; NF-κB, nuclear factor-κB; NO, nitric oxide; OCT4, octamer-binding protein 4; PDCD4, programmed cell death protein 4; PGC1α, peroxisome proliferator-activated receptor-γ coactivator 1α; PPARα, peroxisome proliferator-activated receptor-α; PRMT, protein arginine methyltransferase; SHANK2, SH3 and multiple ankyrin repeat domains protein 2; SREBP, sterol regulatory element-binding protein; TGFβ, transforming growth factor-β; TSP1, thrombospondin 1; UPF1, upstream frameshift 1 regulator of nonsense transcripts 1; ZEB1, zinc finger E-box-binding homeobox 1. [a]Potential liability.

Type I and III interferon response
Distinct interferon cytokine and signalling pathways where by type I (IFNα, IFNβ and others) and type III (IFNλ) interferons are thought to confer potent and tonic front-line inflammatory responses.

Inhibition of PRMT7 was shown to compromise restoration of proteostasis upon treatment with bortezomib, a proteasome inhibitor[82].

Immuno-oncology approaches to treat cancer offer a new therapeutic avenue, and arginine methylation has been reported to play roles in immune regulation (further discussed in the section PRMT inhibitor clinical development). Several studies outline how arginine methylation regulates key transcription factors or splicing programmes in immune cells. FOXP3, a transcription factor critical for regulatory T (T_{reg}) cell identity and function in immunosuppression is dimethylated on R48 and R51 by PRMT5, and possibly by PRMT1 or PRMT6 (reviewed elsewhere[84]). This methylation attenuates the expression of immunosuppressive genes, and can lead to autoimmunity, tumour shrinkage and the associated $CD8^+$ T cell infiltration[85,86]. Concurrently, T cell activation results in a global increase in arginine methylation through upregulation of PRMT5, histone methylation, altered STAT1 signalling and cytokine production[87]. A recent study revealed that symmetrical arginine dimethylation was necessary for the type I and type III interferon response upon T cell receptor stimulation[88]. The authors stipulated that such suppression of type I and type III interferons may be detrimental to the immune response, but they also pointed out studies indicating that a type II interferon response dominates in tumours[89]. As a result, PRMT5 inhibition, which preferentially leads to a type II interferon response, can result in tumour-selective priming of the microenvironment and a therapeutic response to a subsequent immune checkpoint blockade. This extensive disruption of type I and III interferon programmes by blocking symmetrical arginine methylation could not be attributed to regulation of a single factor but was correlated with mRNA splicing alterations affecting specific signalling pathways[88]. Further research is needed to identify the methylation events leading to modulation of the splicing network as it will provide deeper understanding of PRMT5 in immuno-oncology and potential new targets.

Overall, a wealth of PRMT substrates play roles in oncogenic transcription, splicing, signalling and metabolic regulation with promising potential for therapeutic modulation. Nevertheless, a full mechanistic understanding of how arginine methylation and its inhibition influences the cancer phenotype is lacking for the majority of these substrate proteins. Targeting splicing deficiencies and the cofactor SAM (see BOX 1 and the section PRMT inhibitor clinical development) are the approaches most commonly being explored. Furthermore, the immuno-oncology implications of PRMT inhibition are currently under investigation with, for example, clinical trials combining immune response modulators with PRMT5 inhibition. However, further research into immune cell-specific PRMT functions is needed to enhance our mechanistic understanding of arginine methylation in immuno-oncology.

Neurodegenerative diseases. Neurodegenerative diseases such as amyotropic lateral sclerosis (ALS), Huntington disease (HD), Alzheimer disease (AD) and frontotemporal lobar degeneration (FTLD) are often characterized by the accumulation of protein aggregates that include RBPs. Arginine methylation is prevalent in RBPs that are heavily involved in regulating the dynamics of normal physiological phase separation and its pathological irreversible amyloid aggregate accumulation.

The RBP FUS, which has about 20 RG/RGG methylated repeats, regulates RNA splicing, transport, transcription and the DNA damage response[90]. The RG/RGG motifs of FUS reside in the N terminus, which interacts extensively with the carboxy terminus (a prion-like, tyrosine-rich disordered region) of the neighbouring FUS protein, thus eliciting phase separation through the formation of an extensive network of cation–π interactions in a multiprotein manner. PRMT1 methylation of RG/RGG motifs increases the protein density threshold required for phase separation by weakening intermolecular hydrogen bonding, thus regulating phase separation behaviour[91,92].

A large group of heterogeneous nuclear ribonucleoproteins (hnRNPs) are also heavily arginine methylated and show phase separation behaviour. The dynamic stress granule response involving hnRNPs highlights the reversible physiological stress-inducible nature of RNA–protein aggregates; these responses ensure survival when cells encounter harmful stimuli. Conversely, stable pathological aggregates containing hnRNPs have been reported in ALS and FTLD[93]. In ALS, hnRNPA2 mutations increase phase separation of this protein with TAR DNA-binding protein 43 (TDP43), another arginine methylated protein, and, as in the case of FUS, arginine methylation increases the threshold for phase separation to occur[93,94]. As arginine methylation can potentially prevent pathological protein association, increasing the levels of arginine methylation by modulating demethylation could be desirable in ALS. There are very few studies on arginine demethylation. JMJD6 and some other members of the JMJC family of oxygen-dependent hydroxylases are the only proteins reported to demethylate arginines[95,96]; however, a recent study could not provide clear evidence of arginine demethylation activity[97]. Using a generic methylation inhibitor, adenosine dialdehyde (Adox), or PRMT5/PRMT1 knock-down, arginine methylation of Ras GTPase-activating protein-binding protein 1 (G3BP1), a stress granule marker protein, was reduced, leading to increased formation of stress granules[98]. Conversely, JMJD6 depletion or the generic demethylation inhibitor *N*-oxalylglycine impaired stress granule formation and modulated arginine methylation of G3BP1 (REFS[98,99]). It should be noted that, apart from the LSD1 inhibitors that are currently in clinical testing, demethylase modulator development has been challenging[100]. Although use of the generic inhibitor of methylation, Adox (BOX 1), has historically provided valuable data, the results should be interpreted with caution because of the widespread methylation block and toxicity it produces.

Huntingtin protein (HTT) — the well-known mutant form of which has expanded polyglutamine repeats and causes HD — co-localizes with PRMT5. Wild-type HTT recruits PRMT5 to chromatin, enabling H4R3 symmetrical dimethylation and transcriptional repression. The mutant HTT protein has reduced PRMT5 recruitment

3′ Stem–loop melting
A conformational switch facilitated by proteins or RNA that leads to destabilization of stem–loop RNA structures in 3′ regions.

and associated silencing[101]. In addition, it has been increasingly evident that the DNA damage response pathway plays a significant role in HTT pathology[102]. Although several PRMTs have been implicated in positive and negative regulation of the DNA damage response[3,19,32], their exact role, if any, in HD pathogenesis or the potential benefit of PRMT targeting in this disease is not clear. Oxidative stress is linked to the DNA damage response in other neurodegenerative diseases such as AD. PRMT5 and its methylated splicing factors bind the molybdopterin synthase associating complex, which regulates sulfur amino acid catabolism, thus preventing oxidative damage and, potentially, the onset of AD[103].

It has also been reported that PRMT5 protects neuronal cells against amyloid-induced toxicity[104]. The neuroprotective role of most PRMTs and the role of arginine methylation in increasing the threshold of phase separation behaviour suggest that inhibiting PRMTs could potentially exacerbate or promote the development of neurodegenerative diseases. Additionally, the currently available inhibitors of type I PRMTs (MS023 and GSK3368715; TABLE 2) potently affect the neuron-specific and essential PRMT8, suggesting potential off-target effects and highlighting the need for more selective molecules. Thus, further studies are needed to address the neurological consequences of selective PRMT inhibition when blood–brain barrier permeable compounds are used.

Viral response and inflammation. Several viral proteins are methylated by PRMTs, perhaps because they have high RG/RGG content and roles in RNA binding and chromatin regulation. For example, Epstein–Barr virus nuclear antigen 2 (EBNA2), the helicase domain of non-structural protein 3 (NS3) of hepatitis C virus, ICP27 of herpes simplex virus 1, adenovirus E1B-AP5 and HIV transactivator protein (TAT) are all arginine methylated[105–109]. In general, arginine methylation positively regulates viral processes, and inhibition of PRMTs could therefore have antiviral potential. For example, methylation of AU-rich element-binding protein (AUF1/hnRNPD) by PRMT1 is necessary for 3′ stem–loop melting, an essential process in West Nile virus RNA cyclization and replication, and thus, inhibition of PRMT1 decreased viral replication[110]. Arginine methylation also increases the replication of adenovirus and hepatitis virus[111,112]. PRMT5 methylation of hepatitis B virus core protein enables its accumulation and trafficking[113]. Depletion of PRMT5 and PRMT7 drastically reduced the stability of the key HIV protein VPR, which enables the virus to maintain an infection reservoir in human macrophages[114]. However, not all methylation events promote viral replication and sometimes conflicting evidence calls for more studies. For example, PRMT6 restricts HIV infection by methylating and impairing the function of HIV-1 proteins such as TAT, REV and nucleocapsid protein p7 (REFS[105,115,116]), suggesting that PRMT6 protects against HIV infection. However, PRMT6 also seems to enhance TAT function by increasing its half-life and nucleolar retention[117,118]. Although arginine methylation occurs in several viral proteins and many RBPs that interact with viral RNA, the therapeutic translation in infectious diseases has been difficult owing to the complexity of the systems and, previously, the lack of chemical tools. It is not clear whether arginine methylation can be a viable option for host-mediated viral targeting, which is itself a novel and emerging approach that has generated a limited number of targeted therapies so far[119]. Further investigations and in-depth mechanistic studies enabled by tool compounds in robust preclinical models are needed.

Levels of PRMT1, CARM1, PRMT5 and PRMT6 are elevated in inflammatory diseases such as chronic obstructive pulmonary disease and asthma[120–122]. Arginine methylation of transcription and RNA processing factors, as well as signalling molecules, modulates inflammatory responses[123]. NF-κB subunits can be symmetrically dimethylated by PRMT5, promoting the transcriptional activity of NF-κB[124,125]. NF-κB can also recruit PRMT1, CARM1 and PRMT6 to remodel the chromatin environment and drive pro-inflammatory and fibrosis-associated gene transcription[126–128]. Complex transcriptional cascades are activated in inflammation, which change arginine methylation by increasing PRMT1 levels and modulate the IL-4 response in eosinophils during pulmonary inflammation. Conversely, PRMT1 inhibition suppressed eosinophil infiltration and the inflammatory response[121].

In addition to local effects, a large body of research provides evidence that extracellular systemic circulating asymmetrical dimethylarginine (ADMA) and monomethylarginine (MMA), methylated forms of free arginine produced as a result of protein degradation, are important cardiovascular regulators that aggravate inflammation-associated conditions such as pulmonary hypertension[129,130] (BOX 2). This remarkable association of arginine methylation with pulmonary inflammation and fibrosis could be an untapped source for therapeutic intervention. However, given the small amount of ADMA relative to total arginine in cells and in plasma, it is unclear whether there would be a sufficient therapeutic window for PRMT inhibition. Overall, oncology remains the most promising area for further clinical development.

PRMT chemical and structural biology

The roles PRMT enzymes play in numerous diseases have spurred significant interest in targeting them pharmacologically. Although initial discoveries of PRMT function were and continue to be made using genetic means, the development of several chemical inhibitors has further advanced our understanding and underscored the complementarity of genetic and chemical biology approaches. In addition to providing excellent tool compounds to investigate the biological function of PRMTs, these efforts have also provided a deeper structural and mechanistic understanding of PRMTs, which in turn benefited inhibitor development. Recent efforts by the chemical biology community to develop potent, selective, cell-active compounds with well-defined mechanisms of action have provided tools or chemical probes to investigate biological mechanisms. For chemical probe properties and use, we refer the reader to in-depth discussions[131–133]. Building upon previous

Table 2 | **Cell active PRMT inhibitors**

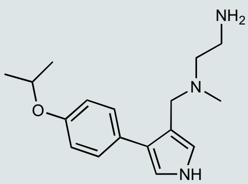
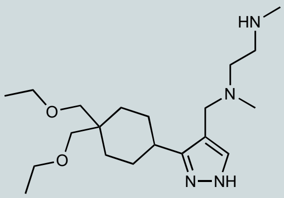
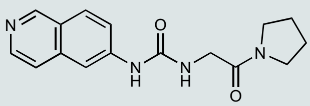
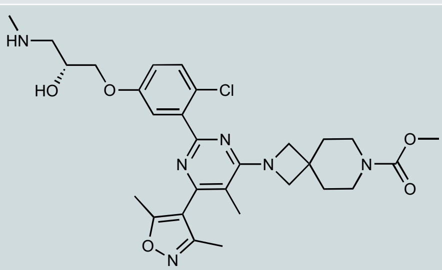
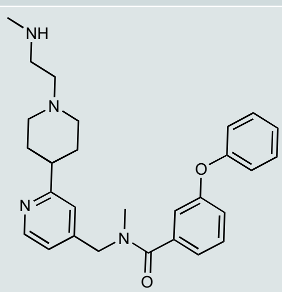
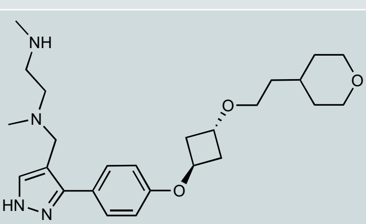
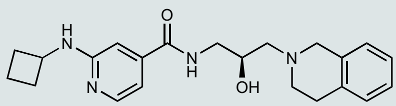
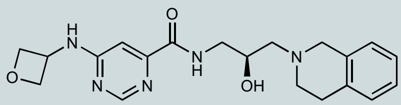
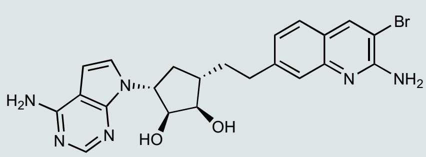
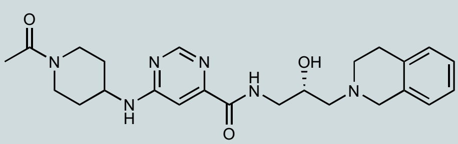
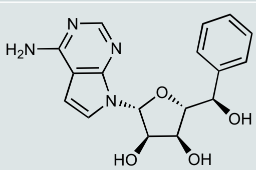
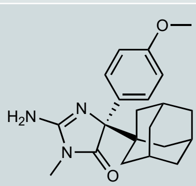
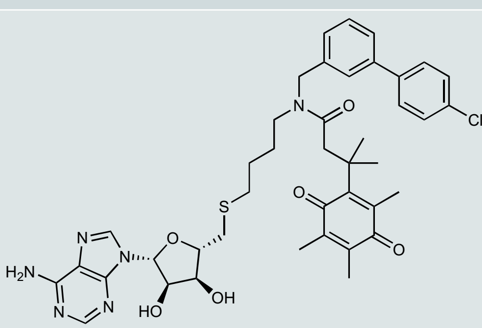
Compound name	Chemical structures	Mechanism of action	PRMT selectivity (in vitro IC_{50})	Cell type and cellular activity	In vivo activity	Year and refs
MS023		Substrate competitive	PRMT1 (30 nM) PRMT3 (119 nM) CARM1 (83 nM) PRMT6 (4 nM) PRMT8 (5 nM)	MCF7 cells (IC_{50} = 9 nM on H4R3me2a; IC_{50} = 56 nM on H3R2me2a)	Antitumour effect in AML PDX mouse models	2016 (REF.[239]) 2019 (REF.[14])
GSK3368715		Substrate competitive	PRMT1 (33.1 nM) PRMT3 (162 nM) CARM1 (38 nM) PRMT6 (4.7 nM) PRMT8 (3.9 nM)	RKO cells (IC_{50} = 13.6 nM on MMA)	Antitumour effect in breast cancer, pancreatic cancer, renal cancer cell lines, xenograft mouse model Antitumour effect in pancreatic cancer PDX mouse model	2019 (REF.[139])
SGC707		Allosteric	PRMT3 (31 nM)	HEK293 cells (91 nM on H4R3me2a)	Impaired hepatic lipid accumulation in a mouse model Induced an osteopenia phenotype in mice Reduced hepatic steatosis in fatty liver disease mouse model	2015 (REF.[155]) 2018 (REF.[188]) 2019 (REF.[44]) 2019 (REF.[240])
EZM2302 (GSK3359088)		Substrate competitive	CARM1 (6 nM)	NCI-H929 cells (IC_{50} = 9 nM on MMA)	Antitumour effect in MM cell line xenograft in mice	2017 (REF.[241]) 2018 (REF.[57])
TP-064		Substrate competitive	CARM1 (<10 nM)	HEK293 cells (43 nM on MED12me2a)	Bioavailable in a mouse model	2018 (REF.[242])
EPZ020411		Substrate competitive	PRMT6 (10 nM)	A375 cells (0.6 μM on H3R2me2a)	Attenuated cisplatin-induced hair cell death in a mouse model	2015 (REF.[151]) 2020 (REF.[243])

Table 2 (cont.) | **Cell active PRMT inhibitors**

Compound name	Chemical structures	Mechanism of action	PRMT selectivity (in vitro IC_{50})	Cell type and cellular activity	In vivo activity	Year and refs
EPZ015866 (GSK591)		Substrate competitive	PRMT5 (4 nM)	NA (in vitro tool compound)	NA	2016 (REF.[244])
GSK3235025 EPZ015666		Substrate competitive	PRMT5 (22 nM)	Z-138 cells (<1 μM on SDMA)	Antitumour effect in MCL, MM, AML, GBM and bladder cell line mouse xenografts Antitumour effect in TNBC PDX mouse model	2016 (REF.[244]) 2015 (REF.[140]) 2018 (REF.[245]) 2018 (REF.[246]) 2018 (REF.[247]) 2018 (REF.[248])
JNJ-64619178		SAM competitive	NA	NA	Antitumour effect in lung cancer, AML, non-Hodgkin lymphoma cell line mouse xenografts	2019 (REF.[249])
GSK3326595		Substrate competitive	PRMT5 (6.2 nM)	Z-138 cells (2.5 nM on SDMA)	Antitumour effect in non-Hodgkin lymphoma cell line mouse xenograft Antitumour effect in DLBCL PDX mouse model	2018 (REF.[69]) 2019 (REF.[174]) 2019 (REF.[250]) 2019 (REF.[176])
LLY-283 (C220)		SAM competitive	PRMT5 (22 nM)	MCF cells (25 nM on SDMA) A375 cells (40 nM on MDM4 splicing)	Reduced acute graft versus host disease incidence in mice Antitumour effect in MPN xenograft	2018 (REF.[146]) 2020 (REF.[177])
Compound1a		Allosteric	PRMT5 (16 nM)	NA	NA	2020 (REF.[153])
SGC3027		SAM competitive	PRMT7 (2.5 nM)	C2C12 cell (2.4 μM on HSP70me1)	NA	–

AML, acute myeloid leukaemia; DLBCL, diffuse large B cell lymphoma; GBM, glioblastoma multiforme; H4R3me2a, histone H4 asymmetrically dimethylated at arginine 3; HSP70, heat shock protein 70; IC_{50}, half-maximal inhibitory concentration; MCL, mantle cell lymphoma; MED12, mediator of RNA polymerase II transcription subunit 12; MM, multiple myeloma; MMA, monomethylarginine; MPN, myeloproliferative neoplasms; NA, not applicable; PDX, patient-derived xenograft; PRMT, protein arginine methyltransferase; SAM, *S*-adenosyl-L-methionine; SDMA, symmetrical dimethylarginine; TNBC, triple-negative breast cancer.

Box 2 | **Circulating methylated arginine species in disease**

Many cardiovascular diseases have been associated with enhanced levels of arginine metabolites such as asymmetrical dimethylarginine (ADMA) and monomethyl-arginine (MMA) (reviewed elsewhere[129]), which are generated by protein arginine methyltransferases (PRMTs). Accumulation of ADMA, an endogenous nitric oxide synthase (NOS) inhibitor, significantly reduces the synthesis of the potent vasodilator nitric oxide (NO) and contributes to cardiovascular disease pathogenesis, including hypercholesterolaemia, stroke, atherosclerosis and coronary heart disease[256–258].

Increased levels of circulating ADMA have also been linked to chronic inflammation and autoimmune diseases such as arthritis, Alzheimer disease, atherosclerosis, asthma, cancer and cardiovascular disease[129,130,259]. Circulating ADMA serves as a biomarker of cardiovascular disease as well as endothelial dysfunction[260,261]. Patients with rheumatoid arthritis not only have higher levels of symmetrical dimethylarginine (SDMA) but also have increased levels of arginase[262]. Arginase locally depletes arginine, and is utilized by myeloid-derived suppressor cells and antigen presenting cells to regulate T cell activation[263].

Both ADMA and SDMA levels are increased in patients with chronic kidney disease, and these metabolites may have a role in the pathogenesis of accelerated atherosclerosis in this population[217,218,264]. Thus, although it may be concluded that reducing circulating ADMA levels by inhibiting type I PRMTs could lead to beneficial outcomes in inflammatory and cardiovascular diseases, more studies are needed. For example, ADMA also has anti-fibrotic effects in the obstructed kidney, suggesting that PRMT inhibitors should be used cautiously in patients with chronic kidney disease[219].

reviews[3,32,134,135], BOX 3 and TABLE 2 provide an overview of a select group of cell-active compounds that meet rigorous potency and selectivity criteria, focusing on their mode of action, structure–function relationship and clinical potential.

Numerous structural studies have illuminated the structural chemistry of PRMT inhibition. Type I PRMTs adopt a head to toe homo-dimeric arrangement, PRMT5 is a homo-tetramer and PRMT7 is a monomer in which adjacent catalytic and pseudo-catalytic domains mimic the homo-dimeric system seen in type I enzymes (FIG. 2). Secondary structural elements of PRMTs that line the catalytic site are dynamic and stabilized upon cofactor and substrate binding, leading to opportunities for allosteric inhibition[136]. Catalysis is believed to be facilitated by a pair of highly conserved glutamic acids involved in the activating deprotonation of the methyl-accepting nitrogen[137,138] (FIG. 2a).

The methyl-donating cofactor SAM and the methyl-accepting arginine occupy distinct but connected binding pockets that can individually be targeted by chemical inhibitors. The affinity of endogenous ligands for each site is an important parameter to consider when selecting a medicinal chemistry strategy, as it will be more challenging to compete with ligands that bind strongly to their target. For example, GSK3368715 (EPZ019997), a type I PRMT inhibitor, occupies the substrate arginine site of PRMT1 (REF.[139]) (FIG. 2a). Similarly, GSK3235025 (EPZ015666), a selective PRMT5 inhibitor and precursor of the clinical compound GSK3326595, directly competes with the substrate arginine[140] (FIG. 2b). Interestingly, a cation–π interaction between the phenyl ring of GSK3235025 and the positively charged sulfonium methyl group of SAM contributes significantly to the binding affinity of the inhibitor. A similar SAM-dependent mode of action was later revealed for other methyltransferase inhibitors, confirming that cofactor binding not only helps stabilize the arginine binding pocket but also offers a vector to increase the binding potency of substrate competitors via direct interaction with SAM[141].

Drug-like ligands can also compete with the cofactor. Although SAM is the cofactor for all protein methyltransferases, SAM mimetics can be surprisingly specific, which probably reflects the structural diversity of the cofactor binding pocket[142]. This was first illustrated by the unexpected specificity of the endogenous metabolite methylthioadenosine (MTA), a SAM analogue that occupies the cofactor binding site of PRMT5 (PRMT5 half-maximal inhibitory concentration (IC_{50}) = 4.6 µM versus >100 µM for other methyltransferases)[143] (FIG. 2b). Methylthioadenosine phosphorylase (MTAP), the enzyme that metabolizes MTA (BOX 1), is frequently deleted in cancer. Mechanistically, MTAP loss results in the accumulation of its metabolite, MTA, which selectively inhibits PRMT5 activity in a SAM competitive fashion, leading to reduced PRMT5 activity and potentially increased sensitivity to PRMT5 inhibition[144,145]. PRMT5 inhibitors that make stabilizing interactions with MTA could potentially be selective for the MTA-bound state of PRMT5 and exploit the PRMT5 dependency of MTAP-deleted cancer cells without inhibiting PRMT5 in MTAP-expressing healthy cells. This strategy is further supported by the structure of PRMT5 bound to MTA and a substrate peptide[143], showing that the substrate arginine occupies a druggable pocket where inhibitors could directly engage MTA (FIG. 2b).

The chemical inhibitor LLY-283, an orally bioavailable SAM analogue, inhibits PRMT5 with low nanomolar IC_{50} and has no activity towards 31 other methyltransferases[146]. Although it is now clear that some SAM analogues can inhibit particular protein methyltransferases with high specificity, the selectivity profile of these compounds across the multitude of proteins that bind adenosine-containing metabolites other than SAM remains to be established. Another challenge in developing cofactor competitors is that the cofactor binding pocket is highly charged, especially around the amino acid group of SAM, and inhibitors need to be sufficiently polar to occupy this binding site with high affinity while retaining favourable physicochemical properties and potency. Bi-substrate inhibitors can sometimes be used to mitigate both selectivity and cell-penetrance issues of SAM analogues. These compounds partially or fully occupy the cofactor site and extend into the arginine binding pocket. For example, SKI-72 is a CARM1 inhibitor that directly competes with both SAM and the substrate arginine, is moderately selective (CARM1 IC_{50} = 43 nM; >10-fold selectivity over 33 other methyltransferases) and recapitulates the effect of *CARM1* genetic deletion in reducing breast cancer cell invasion[147] (FIG. 2a). SGC8158, a bi-substrate inhibitor of PRMT7, is more potent and selective than SKI-72 (IC_{50} < 3 nM; >40-fold selectivity over other methyltransferases)[82] (FIG. 2c). In both cases, the extension from the adenosine scaffold to the arginine binding pocket was necessary to achieve sufficient potency and specificity, but both compounds retained physicochemical liabilities and prodrugs (SKI-73 for CARM1; SGC3027 for PRMT7) were used in cells. A recently reported SAM analogue selectively and potently inhibits PRMT5 (REF.[148]). Like LLY-283 or the clinical PRMT5 inhibitor JNJ-64619178, this compound lacks the methionine moiety of the cofactor, but, unlike LLY-283, it extends slightly into the substrate arginine binding site and forms a hydrogen bond with a catalytic glutamic acid (E444). The inhibitor has good physicochemical properties, is active in cells and provides a more favourable template for further development.

The presence of a cysteine at the ligand binding sites of some PRMTs recently led to the discovery of covalent substrate and cofactor competitors of PRMT6 (FIG. 2a) and PRMT5, respectively[149,150]. However, these compounds were not substantially more potent or selective than previously reported reversible inhibitors[151,152]. Molecules with an allosteric mode of action may be more promising. Binding of the PRMT5 inhibitor compound 1a or the PRMT6 inhibitor SGC6870 to a pocket that is in the vicinity of the substrate binding site induces conformational remodelling leading to the repositioning of one of the two conserved catalytic glutamic acids away from the active site[153,154] (FIG. 2d). A different allosteric mechanism was reported for PRMT3, in which the inhibitor SGC707 occupies a pocket that is close to the active site of a second PRMT3 chain in the context of the functional homo-dimer[155] (FIG. 2d). The inhibitor is buttressed against a helix of the homo-dimeric partner that shapes the substrate binding site and is believed to induce conformational strains that antagonize proper substrate binding. In both cases, the pocket is occluded by side chains in the absence of ligand, making these mechanisms of action unpredictable. Profiling of the PRMT3 and PRMT6 allosteric inhibitors revealed exquisite selectivity. Whether similar or novel allosteric inhibitory mechanisms can apply to other PRMTs is an open question. In particular, an allosteric targeting strategy for PRMT1 would be attractive, as it has so far been challenging to identify a selective inhibitor to this enzyme.

Box 3 | **PRMT tool compounds that paved the way for clinical compounds**

The very first inhibitor of type I protein arginine methyltransferases (PRMTs), arginine methylation inhibitor 1 (AMI-1), is a symmetrical sulfonated urea salt compound, and was identified in a high-throughput screen (HTS) before PRMTs were well understood[265]. Another medium-throughput screen identified a PRMT3 inhibitor that, following extensive optimization, rendered an allosteric, selective and potent inhibitor, SGC707 (REF.[155]). Numerous HTSs and optimization campaigns at GSK and Epizyme resulted in the potent PRMT5 inhibitor GSK3235025 (REF.[140]), which was subsequently optimized to the clinical compound GSK3526595 that is currently in three clinical trials (NCT02783300, NCT03614728, NCT04676516). These campaigns also generated protein structure knowledge that facilitated the development of the PRMT6 inhibitor EPZ020411 (100-fold selectivity for PRMT 6/8/1 over other PRMTs)[151] and the CARM1 inhibitor EZM2302 (100-fold selective over other PRMTs)[241]. Separate screening efforts yielded TP-064, another selective CARM1 compound[242]. Inspired by these advancements in understanding the structure–activity relationships in PRMTs, the type I inhibitor MS023 was developed (half-maximal inhibitory concentration (IC_{50}) = 4–120 nM for PRMT1, PRMT3, PRMT4, PRMT6, PRMT8)[239]. Another type I PRMT compound, GSK3368715, which has a similar selectivity profile (IC_{50} = 3–160 nM for PRMT1, PRMT3, PRMT4, PRMT6, PRMT8)[139] is also in clinical trial (NCT03666988). Discovery of the *S*-adenosyl-L-methionine (SAM) competitive adenosine moiety-based PRMT5 inhibitor, LLY-283 (REF.[146]), started a new generation of PRMT5 inhibitors with this mode of action: two of these compounds, JNJ64619178 (NCT03573310) and PF06939999 (NCT03854227), are in clinical trials.

PRMT inhibitor clinical development

Although targeting PRMTs is still in its infancy, the promising preclinical work discussed above established a rationale for translating some of the PRMT5 and type I PRMTs findings in oncology into clinical trials. So far, inhibitors targeting PRMT5 and type I PRMTs have entered clinical development for patients with haematological malignancies or advanced solid tumours (TABLE 3). GSK3326595, a potent and selective PRMT5 inhibitor, was the first PRMT inhibitor tested in patients. This phase I trial (NCT02783300) in adult subjects with relapsed/refractory solid tumours and non-Hodgkin lymphoma has shown an encouraging clinical response, particularly in bladder cancer and adenoid cystic carcinoma[156]. Moderate adverse effects, such as nausea, anaemia and fatigue, were reported by patients in this trial[156]. The trial is ongoing, and combination with the immunotherapy agent pembrolizumab is planned as part of this trial. Another GSK3326595 phase I/II study (NCT03614728) was initiated to further define the drug safety profile and to assess the clinical response in patients with MDS, chronic myelomonocytic leukaemia (CMML) and acute myeloid leukaemia (AML) (NCT03614728). The phase II part of this evaluation includes a comparison with best available care and the combination of GSK3326595 plus 5-azaciditine in newly diagnosed patients with MDS. The phase II trial for early-stage breast cancer (NCT04676516) has been announced but is not yet recruiting.

Three other PRMT5 inhibitors — JNJ-64619178, PF-06939999 and PRT543 — are also in phase I clinical development to assess their safety, pharmacokinetics and pharmacodynamics and to find optimal doses and schedules in patients with solid tumours and various haematological malignancies (TABLE 3).

The striking antitumour efficacy of the type I PRMT inhibitor GSK3368715 in preclinical models, especially those with MTAP deletion (see below)[139], has established the rationale for its phase I clinical trials in patients with diffuse large B cell lymphoma (DLBCL) and selected solid tumours (NCT03666988). In this trial, the pharmacokinetics, pharmacodynamics and preliminary clinical activity of GSK33368715 in relapsed/refractory DLBCL and select solid tumours with MTAP deletion are being evaluated.

In addition to GSK3368715, which may have preferential activity in MTAP-deleted tumours, compounds targeting methionine adenosyltransferase (MAT2A), the enzyme responsible for SAM biosynthesis, have been efficacious in MTAP-null tumours[157]. PRMT5 activity depends on MAT2A function, especially in the high-MTA environment of MTAP-deficient cells (see BOX 1), as PRMT5 has low affinity for SAM but high affinity for MTA[145]. In a phase I trial (NCT03435250), the MAT2A inhibitor AG-270 showed the desired pharmacokinetic and pharmacodynamic effects but also significant toxicity and no clear efficacy. The trial is now assessing AG-270 in combination with taxane-based regimens for non-small cell lung and pancreatic cancer. Given that patient selection is based on the MTAP deletion that preferentially seems to affect PRMT5, further clinical testing will reveal the effectiveness of targeting the MTA2A–MTAP–PRMT5 axis.

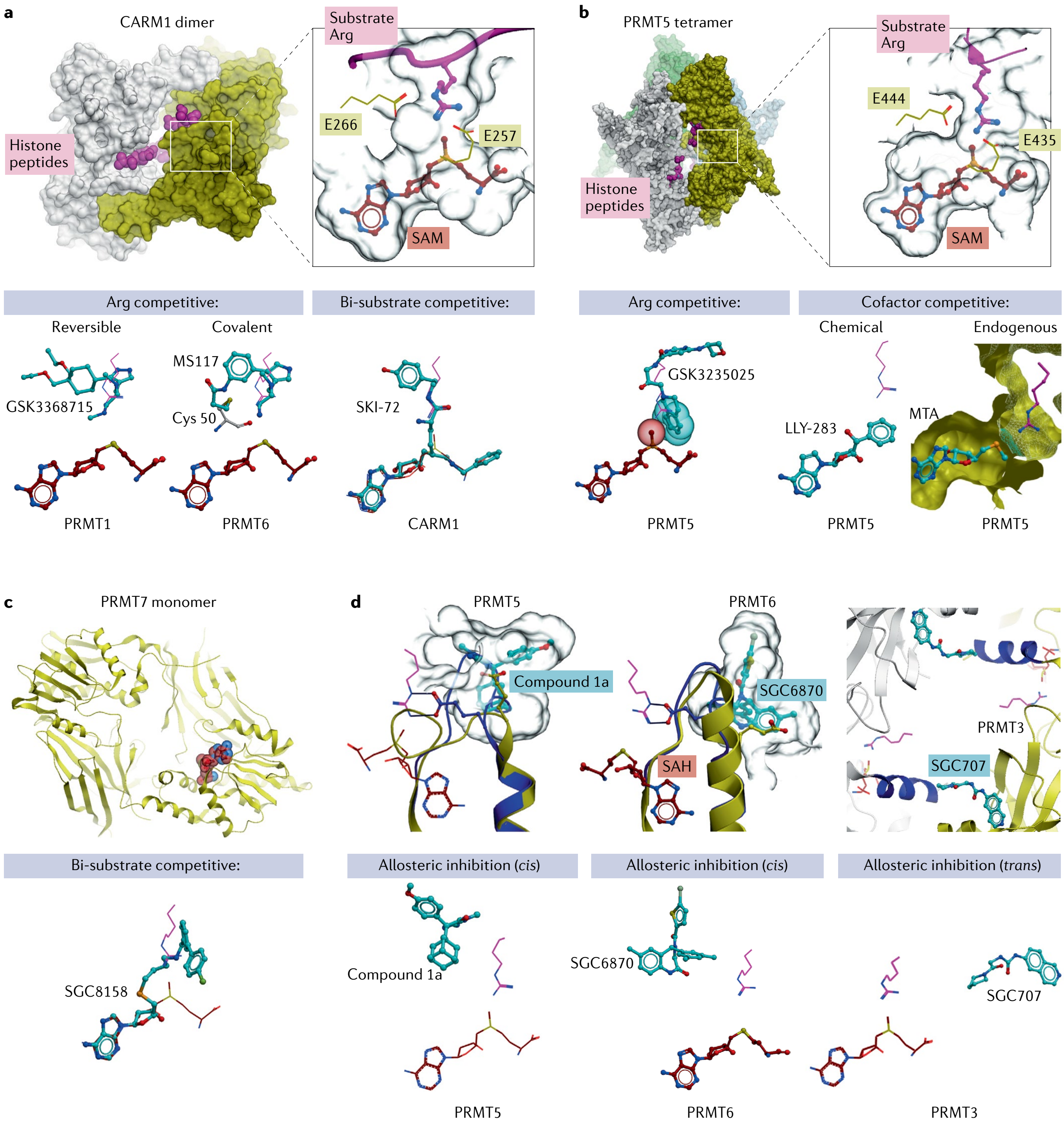


Fig. 2 | **Structural mechanism of PRMT inhibition. a** | Type I protein arginine methyltransferases (PRMTs), such as CARM1, are homo-dimers (top, white and yellow). The substrate peptide (pink) and cofactor *S*-adenosyl-L-methionine (SAM; maroon) occupy distinct but connected binding sites that include two conserved catalytic glutamic acids (inset). Binding poses for the substrate competitors GSK3368715 (reversible binding) and MS117 (covalent binding) as well as the bi-substrate inhibitor SKI-72 are shown (with SAM or SAH, where applicable) bound to PRMT1, PRMT6 and CARM1, respectively. **b** | PRMT5 is a homo-tetramer (top). PRMT5 substrate competitors such as GSK3235025 (EPZ015666) make direct interaction with the departing methyl group of SAM (spheres). LLY-283 and methylthioadenosine (MTA) occupy the cofactor binding pocket. PRMT5 inhibitors exploiting the substrate arginine binding site could directly engage MTA in an MTA-dependent binding mode. **c** | PRMT7 monomer mimics the dimeric architecture of type 1 PRMTs. SGC8158 is a bi-substrate PRMT7 inhibitor. **d** | Allosteric inhibitors of PRMT5 (compound 1a) and PRMT6 (SGC6870) induce conformational remodelling that dramatically shifts a catalytic glutamic acid from a catalytically competent (dark blue) to an inactive (yellow) state (left, middle). The allosteric PRMT3 inhibitor SGC707 induces conformational strains of an α-helix of the homo-dimeric partner (dark blue) that alters the structure of the catalytic site (right). When absent from a structure, the expected pose of the cofactor or arginine substrate are shown as maroon or pink wire traces, respectively. PDB codes: CARM1 dimer [5DX0], GSK3368715 [6NT2], MS117 [6P7I], SKI-72 [6D2L], PRMT5 tetramer [4GQB], EPZ015666 [4X61], LLY-283 [6CKC], MTA [5FA5], PRMT7-SAM [4C4A], SGC8158 [6OGN], PRMT6 active state (blue) [4Y30], compound 1a [6UXY], SGC6870 [6W6D], SGC707 [4RYL].

Table 3 | **PRMT inhibitors in clinical trials**

Target	Name of drug	Company or sponsor	Disease	Phase and status	ClinicalTrials.gov identifier	Study start date
Type I PRMTs	GSK3368715	GlaxoSmithKline	Solid tumours, DLBCL	I (recruiting)	NCT03666988	September 2018
PRMT5	JNJ-64619178	Janssen Research & Development, LLC	Advanced solid tumours, NHL and lower risk MDS	I (recruiting)	NCT03573310	June 2018
PRMT5	PF-06939999	Pfizer	Advanced or metastatic solid tumours	I (recruiting)	NCT03854227	February 2019
PRMT5	GSK3326595	GlaxoSmithKline	MDS and AML	II (recruiting); combination with 5-azacitidine included in newly diagnosed MDS	NCT03614728	August 2018
			Solid tumours and NHL	I (recruiting); combination with intravenous pembrolizumab included	NCT02783300	August 2016
			Early-stage breast cancer	II (not yet recruiting)	NCT04676516	Not yet recruiting
PRMT5	PRT543	Prelude Therapeutics	Advanced solid tumours and haematological malignancies	I (recruiting)	NCT03886831	March 2019

AML, acute myeloid leukaemia; DLBCL, diffuse large B cell lymphoma; MDS, myelodysplastic syndrome; NHL, non-Hodgkin lymphoma; PRMT, protein arginine methyltransferase.

The promising preclinical and clinical results of PRMT inhibitors discussed above provide evidence of their antitumour potential in a subset of patients with cancer. Key remaining questions related to improving the safety profile, maximizing the therapeutic window and identifying combination regimens must be addressed for continued clinical development. In the rest of this section, we summarize the preclinical rationale for patient selection via biomarkers and combination approaches, which could improve the clinical efficacy of PRMT inhibitors.

Patient selection. The clinical use of PRMT inhibitors should ideally be guided by biomarkers that can identify the population of patients most likely to respond to the therapy. In some cases, the target itself might serve as a patient selection biomarker. For example, in human lymphoma cells and mantle cell lymphoma (MCL) cells, Epstein–Barr virus (EBV) infection promotes PRMT5 overexpression and therefore silences several tumour suppressor genes, such as protein tyrosine phosphatase, that are essential to B cell lymphomagenesis[158]. In vivo mouse models showed that PRMT5 inhibitors are selectively cytotoxic in these tumour subtypes; however, the utility of the EBV biomarker remains to be proved clinically[140].

Loss of certain genes has also been linked to an increased sensitivity to PRMT inhibition. MTAP is homozygously deleted in approximately 15% of all human cancers owing to its proximity to the tumour suppressor p16, which resides in the *CDKN2A* locus[159]. MTAP-deficient cells often have enhanced dependency on PRMT5 activity, likely because of the accumulation of MTA[143–145]. The vulnerability in MTAP-null cells is also seen in cells lacking either the upstream enzyme MAT2A or PRMT5 co-complex members such as RIOK1 (REF.[145]). Thus, targeting the MAT2A–PRMT5–RIOK1 axis is a potential therapeutic strategy to address the unmet clinical need in human cancers with MTAP loss. Type I PRMT inhibition was also synthetically lethal with MTAP deletion in lymphoma, melanoma and pancreatic cancer cell lines, but not all MTAP-deficient tumours (for example, breast cancer) respond to the type I PRMT inhibitor MS023; thus, other determinants of sensitivity are likely to be present[139].

The implication of arginine methylation in RNA splicing has also been documented: leukaemia cells with mutations of the RNA spliceosome component SRSF2 are sensitive to PRMT5 inhibition[14]. These preclinical findings have led to the clinical testing of the PRMT5 inhibitor GSK3326595 in patients with MDS or AML and mutations in splicing factors, such as SF3B1, SRSF2, U2AF1 or ZRSR2 (NCT03614728). The rationale for these studies derives from the high frequency of these mutations in MDS and AML and the potential creation of synthetic lethality in which compromised splicing regulation is further exacerbated by PRMT5 inhibition. Mechanistically, understanding the PRMT5 and PRMT1-dependent intron retention or detention and exon-skipping mechanisms will require further research, particularly to examine the interplay between splicing factors, arginine methylation and functional outcomes. Of note, PRMT5 regulates MDM4 splicing and thus protein stability, as the PRMT5-regulated isoform of MDM4 is stable and would therefore ensure suppression of the tumour suppressor p53 (REF.[27]). Sensitivity to PRMT5 inhibitors has often been observed in cells harbouring wild-type p53 (REF.[69]), but, as with MTAP, the p53 status is not a uniform, universal biomarker of response to PRMT5 inhibition[70].

Several other biomarkers for the responsiveness to PRMT inhibitors are emerging. For example, translocation of a complex containing transmembrane protease serine 2 (TMPRSS2) and ETS transcription factor ERG might predict sensitivity to PRMT5 inhibition in prostate cancer[160]. Mechanistically, TMPRSS2–ERG-positive prostate tumours accumulate ERG, which binds to and recruits PRMT5 to methylate AR. Subsequently, the high methylation level of AR blocks its binding to target genes, possibly promoting the development of prostate

R-loops
Transcription-associated RNA–DNA hybrid structures resulting in the displacement of single-stranded DNA.

cancer[74]. It will be interesting to further understand which PRMT inhibitors would function in prostate cancer cells, and whether this inhibitor can be translated clinically to treat this disease.

Combination approaches. Although PRMT inhibitors have shown efficacy as monotherapies in preclinical mouse models of diverse cancer types, their clinical use will likely be integrated into combination regimens with existing or novel therapies. As such, numerous preclinical studies are ongoing to evaluate the efficacy of PRMT inhibitors with epigenetic modulators, the standard of care, kinase inhibitors and immune-based therapies (TABLE 4).

Multiple epigenetic regulatory pathways are dysfunctional in some cancers. Clinical studies combining a PRMT5 inhibitor (GSK3326595) with a DNA methyltransferase (DNMT) inhibitor (5-azacitidine) are in progress in patients with newly diagnosed MDS (NCT03614728). DNMT is thought to directly bind to the PRMT5-specific H4R3me2s mark to promote DNA methylation. This suggests a potential role for DNMT inhibition in augmenting the effectiveness of PRMT5 inhibitors by re-expressing silenced genes[161].

Table 4 | **Combination therapies with PRMT inhibitors**

Target PRMT (inhibitor)	Combinational treatment target (compound)	Cell lines, disease model and clinical trials
Type I PRMTs (MS023)	Cisplatin	Ovarian cancer cell lines[167]
PRMT5 (GSK3186000)	PARP (olaparib)	AML cell lines[166]
Type I PRMTs (MS023)	PRMT5 (GSK3235025 (EPZ015666))	AML cell lines, xenograft and PDX models[14]; lung and pancreatic cancer cell lines[163]
Type I PRMTs (GSK3368715)	PRMT5 (GSK3203591)	DLBCL cancer cell lines; pancreatic cell line and xenograft[139]
PRMT5 (GSK3326595)	DOT1L (EPZ5676)	MLL rearranged cell line[162]
PRMT5 (GSK3326595)	DNMT (5-azacitidine)	Newly diagnosed MDS (NCT03614728)
PRMT5 (GSK3326595)	PD1 (pembrolizumab)	Solid tumours and NHL (NCT02783300)
PRMT5 (DS-437)	ERBB1 (antibody)	Breast cancer xenograft[86]
PRMT5 (PJ-68)	TKI (imatinib)	CML xenograft[172]
PRMT5 (GSK3326595)	AKT (triciribine)	DLBCL cell lines[176]
PRMT5 (GSK3326595)	CDK4/6 (palbocicib)	Breast, oesophageal and pancreatic cancer cell lines; melanoma cell lines and xenograft[174]
PRMT5 (GSK3326595)	mTOR (pp242)	GBM cell lines and xenograft[175]
PRMT5 (C220)	JAK (ruxolitinib)	MPN cell line and xenograft[251]
PRMT5 (GSK3326595)	EGFR (erlotinib)	TNBC cell lines[250]
PRMT5 (GSK591)	BCL-6 (FX1)	DLBCL cell lines[252]
PRMT1 (furamidine)	EGFR (erlotinib)	NSCLC cell lines[253]

AML, acute myeloid leukaemia; CDK4/6, cyclin-dependent kinase 4/6; CML, chronic myeloid leukaemia; DLBCL, diffuse large B cell lymphoma; DNMT, DNA methyltransferase; DOT1L, disruptor of telomeric silencing 1-like; EGFR, epidermal growth factor receptor; GBM, glioblastoma; MDS, myelodysplastic syndrome; MLL, mixed lineage leukaemia; MPN, myeloproliferative neoplasm; NHL, non-Hodgkin lymphoma; NSCLC, non-small cell lung cancer; PARP, poly(ADP-ribose) polymerase; PD1, programmed cell death protein 1; PDX, patient-derived xenograft; PRMT, protein arginine methyltransferase; TKI, tyrosine kinase inhibitor; TNBC, triple-negative breast cancer.

Other compounds that have been shown to synergize with PRMT inhibitors in MLL-rearranged leukaemia models include a disruptor of telomeric silencing 1-like (DOT1L) inhibitor, but the reasons behind these synergies are not fully understood[162].

A robust combinatorial effect exists between PRMT5 and type I PRMT inhibitors[14,139,163]. Many PRMT5 and PRMT1 substrates are shared[14,139], thus, ablation of methylation by one methyltransferase does not necessarily result in loss of the target methylation. Indeed, loss of PRMT1, which methylates substrates asymmetrically, results in increased levels of MMA and symmetrical dimethylarginine (SDMA)[22], further indicating a functional overlap between arginine methylation states, substrates and potential complex consequences of methylation inhibition. Increased baseline levels of MMA and SDMA have also been associated with MTAP deficiency and splicing defects[14,139,163], but this correlation is not present in all tumour types. Therefore, determining the additional factors needed to identify patients who will benefit from this combination therapy will require further clinical investigations, but this combination could be effective and have reduced toxicity in a subset of patients.

PRMTs play important roles in the maintenance of genome stability and the promotion of DNA damage repair mechanisms (as reviewed elsewhere[32,164]). PRMT5 promotes homologous recombination-mediated DNA repair by regulating a multifunctional complex containing histone acetyltransferase KAT5 (also known as TIP60), which has roles in both histone modification and DNA repair[165,166]. Accordingly, a PRMT5 inhibitor (GSK3186000A) was used to overcome resistance to inhibitors of poly(ADP-ribose) polymerase (PARP), which induce cell death by inhibiting DNA repair in AML cells[166]. PRMT inhibition also impedes DNA damage repair by disrupting DNA-dependent protein kinase (DNA-PK) signalling and interfering with the histone marks that regulate DNA repair; PRMT inhibition thus sensitizes cells to cisplatin[167,168]. Furthermore, epigenetic silencing of genes that encode components of drug efflux pumps (such as ABCG2) has been associated with gemcitabine resistance and can be reversed by PRMT3 inhibition[169]. Both CARM1 and PRMT5 have been implicated in the resolution of R-loops (which form in DNA:RNA hybrids), but whether R-loop-induced DNA damage contributes to the effects of PRMT inhibitors in cancer remains unknown[170]. PRMT inhibitors could therefore be combined with chemotherapy to induce responsiveness or overcome resistance to cytotoxic treatments.

Given the role of arginine methylation in cell signalling, another potential application of therapeutic PRMT modulation is to target resistance to kinase inhibitors or the cancer cell populations, such as cancer stem cells (CSCs), that are drug tolerant[32,171]. PRMT5 regulates proliferation and self-renewal of chronic myeloid leukaemia (CML) stem cells, so a PRMT5 inhibitor (PJ-68) was used to eradicate these CSCs and overcome resistance to tyrosine kinase inhibitors[172]. Likewise, PRMT5 facilitates breast cancer stem cell survival, through epigenetic regulation of the transcription factor FOXP1 or methylation of KLF4 (REFS[59,173]). Thus, breast cancer seems to be an especially promising indication for PRMT5 inhibitors, as is reflected by patient selection cohorts in the clinical trials (TABLE 3). Further understanding of the roles of PRMTs in tumour-specific CSCs and the fate of CSCs upon PRMT inhibition might help to rationally design combination treatment strategies and to translate these findings into the clinic for long-term patient care.

Other compounds that synergize with a PRMT5 inhibitor include an AKT inhibitor in DLBCL cells, an mTOR inhibitor in glioblastoma and a cyclin-dependent kinase 4/6 (CDK4/6) inhibitor in melanoma[174–176]. However, the reasons for these synergistic effects are not fully understood. To date, only inhibitors of PRMT5 have been investigated in combination with kinase inhibitors. Therefore, the extent to which kinase inhibitors would be an effective combination with other PRMT inhibitors remains to be seen.

Emerging preclinical evidence suggests that PRMTs may modulate the transcriptome of tumour and immune cells, driving the interest in combining PRMT inhibitors with immune-based therapies. For example, in the drug-resistant syngeneic murine breast cancer model, a PRMT5 inhibitor (DS-437) decreased the activity of tumour-specific T_{reg} cells in a FOXP3-dependent manner, which prompted antitumour immunity and synergized with anti-ERBB2 monoclonal antibodies[86]. An ongoing phase I clinical trial (NCT02783300) is testing PRMT inhibitors (GSK3326595) in combination with anti-PD1 or anti-OX40 antibodies, and mechanistic aspects of the combinatorial effect are under intense investigation. Intriguingly, recent studies suggest that PRMT5 inhibition could prime the tumour microenvironment for successful immunotherapy by selectively targeting interferon signalling[88,89]. Given the broad effects of PRMTs on transcription in multiple immune subtypes, understanding the role of arginine methylation in these different cells is a precursor to the rational design of combination strategies. It will also be important to identify hypersensitive populations and biomarkers of immune infiltration that predict maximal synergy with immunotherapies.

Opportunities outside oncology

The most advanced PRMT drug development programmes are those for PRMT1 and PRMT5 inhibitors in oncology. In several other areas, such as inflammatory diseases and virology, the role of arginine methylation has recently been validated in animal models, enabled by selective tool compounds or genetic means, thus providing potential areas for therapeutic development.

PRMT inhibitor tool compounds have shown efficacy in inflammatory disease models, including those for pulmonary inflammation, ulcerative colitis, graft versus host disease (GVHD) and multiple sclerosis (using the experimental autoimmune encephalomyelitis model). PRMT5 inhibitors are effective in mouse models of acute GVHD, as these compounds decreased T cell proliferation and cytokine production[177]. In animal studies, the pan type I PRMT inhibitor AMI-1 considerably ameliorates pulmonary inflammation with associated reductions in pro-fibrotic activities and the immune response[121,178]. PRMT1, which has been implicated in the regulation of fibroblasts and pro-inflammatory cytokine production using genetic means, was the intended target of AMI-1 in these studies; however, the selectivity of AMI-1 is not clear, thus, it would be interesting to test whether the potent, bioavailable type I PRMT compound MS023 or its clinical counterpart GSK3368715 (TABLE 2) is efficacious in these models. Similarly, AMI-1 treatment reduced the incidence of colitis and inflammatory bowel disease in a mouse model, possibly by inducing T_{reg} cell differentiation[179], but these studies await confirmation using other preclinical compounds. Thus, continued studies using selective and potent PRMT5 inhibitors and the type I PRMT-targeting compounds may help corroborate these findings and provide a clear rationale for the clinical development of PRMT inhibitors outside oncology.

Conclusions and outlook

Recognition of the role that arginine methylation plays in the epigenetic control of disease development and progression has opened new avenues for drug discovery and therapeutics, and has led to PRMT inhibitor testing in clinical trials for cancer treatment (TABLE 3). However, the mechanisms behind the anticipated therapeutic benefit of PRMT inhibition are not fully understood owing to the complexity of arginine methylation. Given the pleiotropic roles of PRMTs on diverse cellular processes, including transcription, splicing, signal transduction and the DNA damage response, the effects of PRMT inhibition are challenging to predict. In addition, owing to the crosstalk between PRMT proteins, inhibition of one PRMT may have considerable consequences on the activity of another, presenting an extra layer of complexity. Thus, defining the molecular and biological roles of distinct PRMTs and PRMT-containing complexes will enable us to understand the various antitumour activities of PRMT inhibition and to provide mechanistic insights for the selection and stratification of patients who are potential candidates for PRMT inhibitor therapy. The growing collection of preclinical tool compounds that are selective for specific PRMT enzymes or subtypes of enzymes will be useful tools for such studies.

Another challenge when considering PRMTs as therapeutic targets is their fundamental roles in development and normal cellular homeostasis as evidenced by mouse knockout studies, and the role of PRMTs in development and neurodegeneration as discussed above. The breadth of functionality amongst proteins that are methylated at arginine indicates that functional modulation of diverse cellular pathways by PRMTs could have a much broader effect on cellular physiology than originally thought. As such, future studies to identify the full repertoire of PRMT-specific methylated substrates and their downstream cellular functions in both normal and disease-associated cells are required. Regulation of PRMT enzymes themselves at the transcriptional and post-transcriptional levels have been studied in only a limited number of systems. For example, post-translational modification, including methylation, phosphorylation and ubiquitylation, have emerged to regulate the activity, function and stability of PRMTs as well as their protein–protein interactions (as reviewed elsewhere[180]). Future studies on other post-translational modifications, and the enzymes that mediate these modifications, as potential therapeutic targets are needed. Another issue to be considered is that arginines in histones are mutated in some cancer types[181]. These mutations may have a similar oncological function to the oncogenic lysine mutations in histones, such as those that alter H3K27M and H3K36M. A systematic analysis to identify the effects of these mutations on neighbouring histone modifications, PRMT activity and the global gene expression profile should provide a robust mechanistic rationale for the development of personalized medicine for the treatment of tumours with these specific mutations.

The development of potent, selective inhibitors for PRMTs has enabled preclinical investigations in many complex disease systems and paved the way for clinical development. Several mechanisms of action have been exploited, especially for PRMT5 inhibitors, which are the front runners in clinical testing. However, many questions remain. For example, regulation of the levels of the cofactor SAM are complex in normal cells and are likely abnormal in many cancer cells, which necessitates further research to understand the potential efficacy of PRMT inhibitors. As both the SAM-cooperative (that is, does not compete with SAM) PRMT5 inhibitor EPZ105666 and substrate competitive inhibitors failed to induce synthetic lethality in MTAP-deficient cells[143–145], this suggests either that the MTA–PRMT5 regulation in cells is much more complex than previously assumed or that a new mode of PRMT5 inhibition is needed. As such, it will be interesting to see whether MTAP-deleted cells are more sensitive to the recently described allosteric PRMT5 inhibitor or to future compounds that may be selective for the MTA-bound state of PRMT5. Recent work on proteolysis targeting chimera (ProTAC) compounds for PRMT5 (REF.[182]) has provided another path for PRMT modulation that will provide valuable information on testing the utility of PRMT degrader compounds in model systems.

It is increasingly apparent that, in cancer, PRMT inhibitors will be most broadly used in combination with other agents. Preclinical combination studies of PRMT inhibitors with standard chemotherapy agents are already proving to be effective in inducing responsiveness or overcoming resistance to cytotoxic treatments. Haematological malignancies (AML, MDS) with an established rationale of splicing regulation deficiencies are front runners for testing the efficacy and safety of inhibitors of PRMT5 and type I PRMTs. The combination of PRMT5 inhibition with azacitidine, a well-established therapy for AML and MDS, in ongoing clinical trials for MDS (NCT03614728) will test this strategy. It is possible that the synergistic effects of these drugs will vary in different cancers because of the distinct arginine methylation profiles of individual tumours. As such, further mechanistic understanding is required to provide a rationale for their combinational usage in cancers with defined genetic or epigenetic lesions.

In addition to cancer cells, PRMTs can regulate the immune response, including T cell function, antitumour immunity and cytokine signalling. This provides an opportunity to target these immuno-oncology mechanisms. It is interesting to note a recent study of spliceosome targeting therapies in which disruption of mRNA splicing promoted cancer-cell intrinsic antiviral signalling, downstream adaptive immune response and cell death[183]. Given the strong role of PRMTs in splicing regulation, it is worthwhile to investigate whether similar mechanisms may be exploited with PRMT inhibitors, and whether there is potential therapeutic benefit of combining PRMT inhibitors and immunotherapy. Finally, other therapeutic areas such as inflammation await the preclinical validation of PRMTs as therapeutic targets.

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